

Original Article

The effect of Yinhuang Hanhua dropping pill on acute pharyngitis via a systematic “Compound-Target-Pathway” analysis

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ABSTRACT

Background: Yinhuang Hanhua dropping pill (YHHHDP) is a Chinese herbal formula that is commonly used in medical practice for managing of acute pharyngitis (AP) and other upper respiratory tract infections. However, the active components, molecular targets, and mechanisms underlying its therapeutic effects remain poorly understood.

Purpose: The objective of this study was to identify the active components and molecular mechanisms of YHHHDP in the therapy of AP.

Methods: The efficacy of YHHHDP was assessed in an ammonia-induced AP rat model and LPS-stimulated RAW264.7 macrophages based on HE staining, immunofluorescence, ELISA, and qRT-PCR. The active components, molecular targets, and pathways were investigated using integrating compositional analysis, metabolomics, and network pharmacological analysis. Furthermore, the interactions between active components and targets, as well as the key signaling pathways were verified using Western blotting (WB), Bio-layer interferometry (BLI), cellular thermal shift assay (CETSA), etc.

Results: YHHHDP significantly ameliorated the pathological damage and inflammatory response in the pharyngeal tissues of AP rats. YHHHDP also inhibited LPS-induced expression of pro-inflammatory factors in RAW264.7 cells. Research involving component analysis, network pharmacology, and metabolomics showed that baicalin, chlorogenic acid, and 10 other compounds are the active components of YHHHDP used in AP treatment. The core targets of these active ingredients were TNF-α, AKT1, and COX-2. The mechanism of action primarily involved the PI3K-AKT/NF-κB/MAPK signaling pathway. Tryptophan and arachidonic acid metabolism were the primary metabolic pathways. Meanwhile, animal and Cellular verification experiments demonstrated that the active ingredients of YHHHDP can stably bind to key target proteins, thereby regulating key metabolites in tryptophan and arachidonic acid metabolism, inhibiting the expression of key proteins in the PI3K-AKT/NF-κB/MAPK signaling pathway, reducing the release of pro-inflammatory factors, and improving the symptoms of AP.

Conclusion: This study revealed that 10 components, such as baicalin and luteolin, are the active ingredients of YHHHDP in AP treatment. The mechanism of action involves the inhibition of TNF-α, AKT1, and COX-2, thereby exerting therapeutic effects through the regulation of arachidonic acid and tryptophan metabolism and the inhibition of the PI3K-AKT/NF-κB/MAPK signaling pathway. This study provided an invaluable scientific basis for the clinical application and development of YHHHDP.

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Introduction

Acute pharyngitis (AP) is a prevalent upper respiratory tract infection characterized by acute, non-specific inflammation, typically triggered by chemical stimuli or bacterial or viral infections. This condition is characterized by its persistent, seasonal, recurrent, and multifactorial nature, significantly affecting human health and daily life (Huang et al., 2024; Wang et al., 2024). Previous research has demonstrated a strong correlation between the inflammatory response in AP and the metabolic changes that occur naturally *in vivo* (Ding et al., 2023). Nevertheless, the exact mechanisms driving this relationship are not fully understood.

Yinhuang Hanhua Dropping Pill (YHHHDP) is a Chinese medicine compound preparation consisting of *Lonicera japonica* Thunb and *Scutellaria baicalensis* Georgi in a 1:1 ratio. YHHHDP disperses wind, clears heat, and detoxifies (Li et al., 2024b). YHHHDP is mainly prescribed for managing both acute and chronic tonsillitis and pharyngitis and ameliorating infections of the upper respiratory tract (Yan et al., 2021). Although YHHHDP has demonstrated significant clinical efficacy in the management of AP, its effect on the metabolism of AP, and the active ingredients and molecular mechanisms remain unclear.

Combining network pharmacology with experimental studies has gained significant attention in compound preparation of traditional Chinese medicine (TCM) research (Cheng et al., 2024). As a valuable tool, network pharmacology helps identify the specific targets of TCM active ingredients and their interactions within molecular networks (Liu et al., 2024). Meanwhile, metabolomics technology offers a novel perspective on metabolic pathways and key biological processes associated with medications, which has been widely applied in TCM research (Xu et al., 2024). The combination of network pharmacology and metabolomics overcomes the limitations of each approach. Network pharmacology often relies on theoretical predictions without sufficient experimental validation, and metabolomics may fail to provide direct insights into the molecular mechanisms driving observed metabolic alterations. Integrating these fields can enhance our understanding by addressing these shortcomings (Chen et al., 2024).

In this study, YHHHDP was assessed using both rat models and cellular experiments. We conducted a thorough analysis of its active components and primary targets, identified the specific mechanisms related to metabolic disorders through an integrated approach combining network pharmacology and metabolomics, and validated them experimentally *in vitro* from multiple perspectives. This study offer a novel and reproducible research approach to clarify the interactions between various components, targets, and pathways in herbal formulations.

Materials and methods

Reagents

Guizhou Valid Pharmaceutical Co., Ltd. supplied YHHHDP (batch number 20,230,109, Guiyang, China). Aqueous ammonia solution (25–28 %) was purchased from (Sinopharm, Shanghai, China) (20,220,905). The source of dexamethasone (Dex) and lipopolysaccharide (LPS) was Sigma-Aldrich, St. Louis, MO, USA. Rat TNF- α ELISA Kit (ZC-37,624), IL-1 β ELISA Kit (ZC-31,391), IL-6 ELISA Kit (ZC-36,404), and PGE2 ELISA Kit (ZC-37,100) were purchased from ZCIBIO Technology Co., Ltd. Mouse TNF- α ELISA Kit (JL10484), IL-1 β ELISA Kit (JL18442), and IL-6 ELISA Kit (JL20268) were purchased from Shanghai Future Industrial Limited by Share Ltd., Shanghai Sangon Biological Engineering Co., Ltd provided primers for *GAPDH*, *iNOS*, *IL-1 β* , *TNF- α* , and *IL-6*. Chengdu Phytochemical Pure Biotechnology Co., Ltd. supplied baicalin, chlorogenic acid, and ten other compounds. Recombinant human TNF- α , AKT1 protein, was purchased from Sino Biological. Recombinant human COX-2 protein was purchased from MedChemExpress.

Animals and experimental design

All experiments involving animals complied with the NIH guidelines for laboratory animal care and use (NIH Publication No. 80–23; revised 1978), and received approval from the Animal Ethics Committee of Guizhou Medical University (Approval 2022–12–30, Ethics Review No. 2,201,538).

Six to eight-week-old male Sprague-Dawley rats, weighing between 190 and 210 g, were obtained from Beijing Huafukang Biotechnology Co., Ltd. Rats were divided into six groups at random, with each group having eight rats: control, model, Dex (5 mg·kg^{−1}), and YHHHDP at low, medium, and high doses. Based on the principle of clinical dose conversion, the low-dose, medium-dose, and high-dose of YHHHDP were determined to be 0.68g·kg^{−1}, 1.36 g·kg^{−1}, and 2.72 g·kg^{−1}, respectively (Shin et al., 2010). An AP model was developed by applying ammonia spray directly for three consecutive days (Wang et al., 2024). The effectiveness of the model was validated based on decreased food consumption, increased water intake, and noticeable throat inflammation. On day 4, the YHHHDP groups received YHHHDP via gavage once daily for five consecutive days. Dex was given in the positive control group, whereas normal saline was given in the control group. Behavioral and physical parameters (Table S1), including scores for food intake, activity, saliva secretion, mouth hair, and cough variations, were assessed in the ammonia-induced AP rat models after treatment with YHHHDP and Dex.

All rats were anesthetized with an injection of 50 mg/kg of sodium pentobarbital 24 hours after the final dose was given. Subsequently, blood and tissue samples, including those from the pharynx and lungs, were collected for both histopathological and biochemical analyses.

Histopathological analysis

Pharyngeal and lung tissue samples of rats were preserved using 4 % paraformaldehyde, followed by dehydration and paraffin embedding before sectioning. Following staining with hematoxylin and eosin, light microscopy was used to assess histopathological alterations.

Cell experiment

Preparation of serum containing YHHHDP

The two groups of SD rats received either saline or YHHHDP. YHHHDP was administered at a dose of 2.72 g·kg^{−1}. YHHHDP was administered daily for three days. One hour after the final dose, blood samples were obtained from both groups. For further experiments, the supernatant from these samples was centrifuged at 5000 rpm, heated at 56 °C for 30 mins, and then filtered through a 0.22 μ m sterile membrane.

Cell culture

A cell culture medium specifically designed for RAW264.7 cells (ZYPYG0022) was used for the experiment. Cells were used during their logarithmic growth phase, at approximately 80 % confluence, at 37 °C in 5 % CO₂.

Viability assay

The viability of RAW264.7 cells was evaluated through a CCK-8 assay. Cells were seeded in 96-well plates at a density of 2×10^5 cells/ml and left to grow for 24 hours. Concurrently, blank serum or serum containing YHHHDP at final concentrations of 0 %, 2.5 %, 5.0 %, 7.5 %, 10.0 %, 12.5 %, 15.0 %, 17.5 %, and 20.0 % were introduced to the 96-well plates for 24 h. The CCK8 kit was used to evaluate cell viability according to established procedures.

NO assay

RAW264.7 cells were seeded in a 6-well plate and incubated overnight. Thereafter, the cells were divided into six experimental groups: control, model, Dex (50 µg/mL), and three groups with low, medium, and high doses of serum containing YHHHDP. Doses were determined based on cell viability assay. After 6 h of treatment with serum containing YHHHDP or Dex, cells were exposed to LPS (1 µg/mL) for 18 h. A commercial NO assay kit was used to measure NO levels in the culture medium supernatants. Specifically, 100 µL of the supernatant was mixed with 200 µL of reagent 1, and then reagent 2 was added. Afterward, the mixture was centrifuged at 3500 rpm for 15 mins, and then 80 µL of color developer was added to the supernatant. Using sodium nitrite as the standard, absorbance was read at 550 nm to determine nitrite concentration.

ELISA assay

Blood samples were centrifuged at 4 °C for 15 mins, after which ELISA kits were employed to determine the levels of TNF-α, IL-1β et al.

Cells were treated and grouped for the NO assay. After collecting the supernatant, an ELISA kit was used to measure TNF-α, IL-1β, and IL-6 levels in the culture medium.

Quantitative real-time RT-PCR

Total RNA was extracted from cells and converted to cDNA using kits from Promega and TaKaRa. qRT-PCR was then conducted using TaKaRa's TB Green® Premix Ex Taq™ II on a Bio-Rad CFX Connect system. After an initial denaturation at 95 °C for 30 ss, 40 cycles were performed at 95 °C for 5 ss and 60 °C for 30 ss. The sequences of primer are listed in Table S2 (Bao et al., 2019). Using the $2^{-\Delta\Delta CT}$ approach, fold change in relative gene expression was calculated.

Chemical composition analysis of YHHHDP

To extract YHHHDP powdered samples (500 mg), 10 mL of methanol was added, and the mixture was sonicated for 40 mins. A 0.22 µm filter was used to separate the samples after sonication. The analysis was conducted using a Vanquish Horizon UHPLC and a Q-Exactive Plus HRMS from Thermo Fisher Scientific. Gradient elution details are shown in Table S3.

Network pharmacology analysis

The anti-inflammatory active components of two Chinese herbal medicines, *Lonicera japonica* Thunb and *Scutellaria baicalensis* Georgi, were thoroughly analyzed and summarized. These findings were compared with the in vitro chemical composition analysis of YHHHDP. Identified components were analyzed to determine potential targets using the TCMSP and Swiss Target Prediction databases. A search for AP-related targets was conducted in OMIM, GeneCards, DrugBank, and DisGeNET databases using the keyword "AP". The identified targets were integrated with those linked to AP and imported into the Microbiologics bioinformatics platform to identify key targets of YHHHDP for treating AP. The STRING database was then used to analyze these targets for protein-protein interactions (PPI). The KOBAS 3.0 online analysis platform was finally utilized to conduct GO function and KEGG pathway enrichment analyses.

Serum and tissue metabonomics

Samples preparation

A 100 µL aliquot of serum was combined with 400 µL of acetonitrile, vortexed for 1 min, and incubated at -20 °C for 30 mins. The resulting

blend was centrifuged at 12,000 rounds for a duration of 15 mins at 4 °C. The collected supernatant was evaporated using a nitrogen stream and then re-dissolved in 50 % methanol to achieve a final volume of 120 µL.

Pharyngeal tissues from each group were homogenized in ice-cold physiological saline at a weight-to-volume ratio of 1:2 (g/mL). The homogenate was combined with ten volumes of methanol, vortexed, and centrifuged at 12,000 rpm for 10 mins at 4 °C. Using a nitrogen stream, the supernatant was evaporated and redissolved in 120 µL of 50 % methanol.

Detailed information regarding the mass spectrometry injection conditions for sample analysis is presented in Table S4.

Data processing and multivariate statistical analysis

A mzCloud spectrum matching algorithm was used to identify metabolites based on their specific molecular weights and mass spectrometry data collected using Compound Discoverer 3.2 software and an Xcalibur 4.1 workstation. The metabolites were tentatively characterized through comparisons using the METLIN, HMDB, and PubChem databases.

To identify key differences in metabolites, principal component analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA) were executed with SIMCA 13.0 software. Metabolites with variable weight values greater than 1, p-values less than 0.05, and fold changes exceeding 1.2 or below 0.83 were considered significant.

Utilizing the online platform MetaboAnalyst 6.0, differential metabolites were further analyzed for metabolic pathway enrichment. This analysis provided insights into the metabolic pathways significantly associated with the observed differences in metabolites.

Network pharmacology and metabolomics integration

To screen the targets of the identified differential metabolites, the Swiss Target Prediction Database and the Herbal Ingredients' Targets Platform database were utilized. Common targets for the differential metabolites, YHHHDP, and AP were then identified and subjected to PPI and KEGG enrichment analysis. An integrated network of component-metabolite-targets-pathways was developed using Cytoscape 3.8.2, facilitating the identification of the crucial targets and potential pathways of YHHHDP for the treatment of AP. These potential targets were further validated using Western blotting.

Effect of YHHHDP active ingredients on TNF-α-induced activity of L929 cells

The effects of drugs on TNF-α-mediated activity in L929 cells serve as a functional assay for assessing the effects of drugs on TNF-α (Meng et al., 2023). A density of 1.5×10^5 cells/mL was used to seed cells in 96-well plates, allowing them to adhere and grow over a 24-hour period. Subsequently, YHHHDP active ingredients, TNF-α, actinomycin D, and UCB-9260 were added to the wells and incubated 12h. The assay was conducted using an enzyme marker at 450 nm.

Evaluation of AKT1 inhibitory activity of active ingredients of YHHHDP

In the 50µl reaction mixture, there are 40 mM Tris with a pH of 7.4, 10 mM MgCl₂, 0.1 mg/ml BSA, 1 mM DTT, 10 µM ATP, AKT1 Kinase, and the enzyme substrate. The compounds were diluted in 10 % DMSO, and 5µl of this dilution was added to a 50µl reaction mixture. Kinase activity was assessed using the Kinase-Glo Plus luminescence assay kit, which quantifies the ATP remaining in the solution (Luo et al., 2005). The IC₅₀ values were calculated using nonlinear regression with normalized dose–response fit using Prism GraphPad software.

Evaluation of the COX-2 inhibitory activity of active ingredients of YHHHDP

The COX-2 assay buffer's reaction mixture includes COX-2 and the specified quantity of vehicle, reference inhibitor, or test inhibitor. Before adding the substrate arachidonic acid and the detection reagent, the test compounds were incubated for 10 mins. The reaction mixture was then incubated for 5 mins before fluorescence measurement (ex. 535, em. 590). The assay buffer for the negative control did not contain the COX-2 enzyme. Fluorescence signals were measured using a Tecan Infinite M1000 plate reader. Inhibition assays were performed in duplicate at each concentration (Zou et al., 2024).

Molecular docking

The AutoDock Vina software was used to measure the binding affinities between identified components and potential key targets. Binding modes were further analyzed and visualized using PyMOL 2.4.1.

Bio-layer interferometry

Human recombinant proteins TNF- α , AKT1, and COX-2 were dissolved in phosphate buffer solution (PBS) at a concentration adjusted to 200 $\mu\text{g/mL}$ according to the method of a previous study (Zhang et al., 2024); incubated with 10 mM biotin reagent for 1 hour at room temperature, followed by desalting the solution using a desalting column. The biotinylated protein was eluted and collected using PBS. For the BLI experiment, after pre-wetting the streptavidin (SA) sensor, (Sartorius, China) biotinylated TNF- α , AKT1, COX-2 protein was solidified onto the sensor. Various concentrations of the active compounds were then introduced for analysis (Zhang et al., 2021; Zhang et al., 2024). The SA sensors were sequentially equilibrated, bound, and dissociated, and the data was collected and analyzed using Octet BLI Discovery analysis software (Octet®R4, Sartorius, Germany).

Cellular thermal shift assay (CETSA)

The CETSA experiment was conducted following the manufacturer's instructions (Tu et al., 2023). Briefly, after cell collection, cellular proteins were extracted by repeated freeze-thawing with liquid nitrogen. The proteins were then incubated with the active ingredient or DMSO for 1 hour. Thereafter, the proteins were heated at a specific temperature for 3 mins and centrifuged to collect the supernatant. After centrifuging the soluble supernatant for 20 mins, it was used for Western blotting.

Western blotting

Western blotting was performed to detect the expression of relevant proteins in pharyngeal tissues and RAW264.7 cells. The Epizyme BCA assay kit, sourced from Shanghai, China, was used to measure protein concentration in the test samples. Samples were separated by SDS-PAGE and subsequently transferred to PVDF membranes. Following an overnight incubation at 4 °C with the primary antibody solution, the membrane received horseradish peroxidase (HRP)-conjugated secondary antibodies. A chemiluminescence system (ECL) was subsequently used to identify the signal. The band intensities were quantified using ImageJ 1.53 software. Table S5 provides the list of antibodies used in the research.

Statistical analysis

For each experiment, data are shown as mean $\pm$ SEM of a minimum of three biological replicates. SPSS 23.0 and GraphPad Prism 9.5.0 were used for statistical analysis, and $P < 0.05$ was deemed statistically significant.

Results

YHHHDP attenuated ammonia-induced inflammatory response in rats

Behavioral and physical scoring were applied to validate the successful induction of the AP rat model. After modeling, the frequency of mouth scratching and coughing increased, while food intake decreased (see Fig. S1A-E for details).

To assess the potential of YHHHDP in ameliorating pharyngeal tissue damage in rats, we conducted a histological examination using hematoxylin and eosin (H&E) staining. The staining results revealed mucosal lesions in the pharyngeal tissues of model rats (Fig. 1A), including pharyngeal mucosal epithelial deletion (green arrow), necrosis of the lamina propria (black arrow), fibrotic tissue hyperplasia (red arrow), and extensive lymphocyte infiltration. The Dex and different doses of YHHHDP groups exhibited significant improvements in the pharyngeal tissue pathology of the rat AP model compared to the model group. Accumulating evidence supports the beneficial effects of YHHHDP on upper respiratory tract inflammation. TUNEL analysis of pharyngeal tissue cells showed that the AP group had significantly more TUNEL-positive cells than the control group (Fig. S1F). Treatment with Dex and various doses of YHHHDP significantly decreased this positive staining, suggesting the protective effect of YHHHDP against cell apoptosis. Furthermore, we observed that AP damaged the lung tissues of rats. In lung tissues stained with H&E, YHHHDP significantly ameliorated histopathological damage (Fig. S1G).

Inflammatory cytokines, like TNF- α , IL-6, IL-1 β , and PGE2, were measured in rat serum using an ELISA kit. The levels of these cytokines were significantly higher in model rats compared to control rats. Dex and YHHHDP reduced the serum levels of TNF- α , IL-6, IL-1 β , and PGE2, with YHHHDP (2.72 g/kg) exerting the most significant effect (Fig. 1B). The results indicate that YHHHDP has a healing impact on rats suffering from AP.

LPS-induced inflammation can be suppressed by YHHHDP in RAW264.7 cells

First, we measured the cytotoxicity of blank serum and serum containing YHHHDP. No cytotoxic effects were observed on RAW264.7 cells after 24 hours of treatment with serum concentrations ranging from 0.0 % to 20.0 % (Fig. S2A-B). Furthermore, cell viability in the 7.5 %, 10 %, and 12.5 % serum containing YHHHDP groups was comparable to the control group. Therefore, serum containing YHHHDP was categorized into low (7.5 %), medium (10.0 %), and high (12.5 %) concentrations for the following experiments.

We investigated the effects of serum containing YHHHDP on the morphological changes of RAW264.7 cells (Fig. 1C). LPS stimulation induced distinct morphological changes and induced the characteristics of macrophage activation, including cytoplasmic extensions and increased cellular size. Microscopic examination exhibited that serum containing YHHHDP markedly mitigated the morphological alterations induced by LPS in RAW264.7 cells.

LPS stimulation of RAW264.7 cells induces a significant inflammatory response, characterized by NO release. This research evaluated the impact of serum with YHHHDP on NO production induced by LPS in RAW264.7 cells. Fig. 1D shows that serum containing YHHHDP dose-dependently inhibited NO production in RAW264.7 cells in response to LPS. Furthermore, the levels of TNF- α , IL-6, and IL-1 β were reduced in a dose-dependent manner in the groups treated with serum containing YHHHDP.

The expression levels of genes associated with inflammation were measured using RT-qPCR to determine the effects of serum containing YHHHDP on the expression of *iNOS*, *TNF- α* , *IL-6*, and *IL-1 β* . Serum containing YHHHDP significantly reduced the expression of *iNOS*, *TNF- α* , *IL-6*, and *IL-1 β* mRNA after treatment with LPS (Fig. 1E).

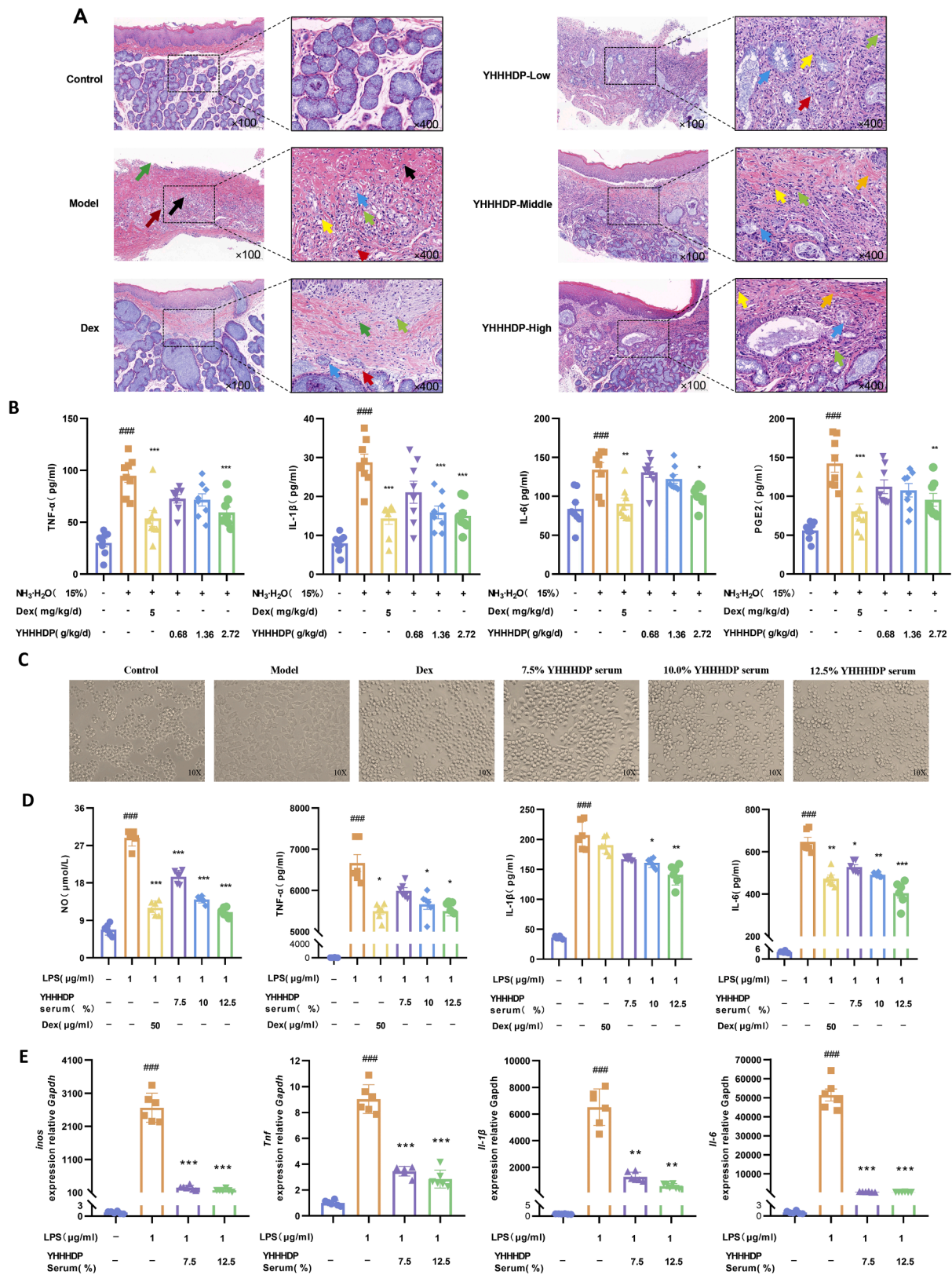


Fig. 1. Therapeutic effect of YHHHDP on ammonia-induced acute pharyngitis in rats and LPS-stimulated RAW 264.7 cells: (A) H&E staining for inflammatory injury of pharynx tissues of rats (Magnification 400 $\times$) ($n = 8$). (B) Effects of YHHHDP on serum TNF- α , IL-1 β , IL-6 and PGE2 (mean $\pm$ SEM, $n = 8$). (C) Effects of the serum containing YHHHDP on the general morphology of the cells (10 $\times$). (D) Effects of serum containing YHHHDP on NO production and pro-inflammatory cytokines including TNF- α , IL-1 β and IL-6 (mean $\pm$ SEM, $n = 6$). (E) The mRNA levels of iNOS, TNF- α , IL-1 β and IL-6 (mean $\pm$ SEM, $n = 6$). Note: ### $p < 0.001$, compared with control group; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, compared with model group.

Predicting of potential active ingredients and pathways for YHHHDP

A network pharmacology approach combined with UHPLC-Q-Exactive Plus MS analysis was utilized to identify potential active components of YHHHDP.

The chemical components of YHHHDP were analyzed using UHPLC-Q-Exactive Plus MS, and the resulting chromatogram is presented in Fig. S3A and Fig. S3B. In total, 71 chemical components were identified (Table S6). The TCMSP database revealed 379 components in *Lonicera japonica* Thunb and *Scutellaria baicalensis* Georgi. Based on a literature review and analysis of anti-inflammatory components reported for these herbs (Table S7), 12 active components of YHHHDP were identified, including baicalin, baicalein, scutellarein, wogonoside, wogonin, galuteolin, chlorogenic acid, luteolin, kaempferol, neochlorogenic acid, isochlorogenic acid B, and cryptochlorogenic acid.

A network pharmacology approach was employed to elucidate the potential mechanism of YHHHDP in treating AP after identifying its active components. The network analysis revealed that the 12 active components targeted 212 drug targets, and 2329 disease-related targets

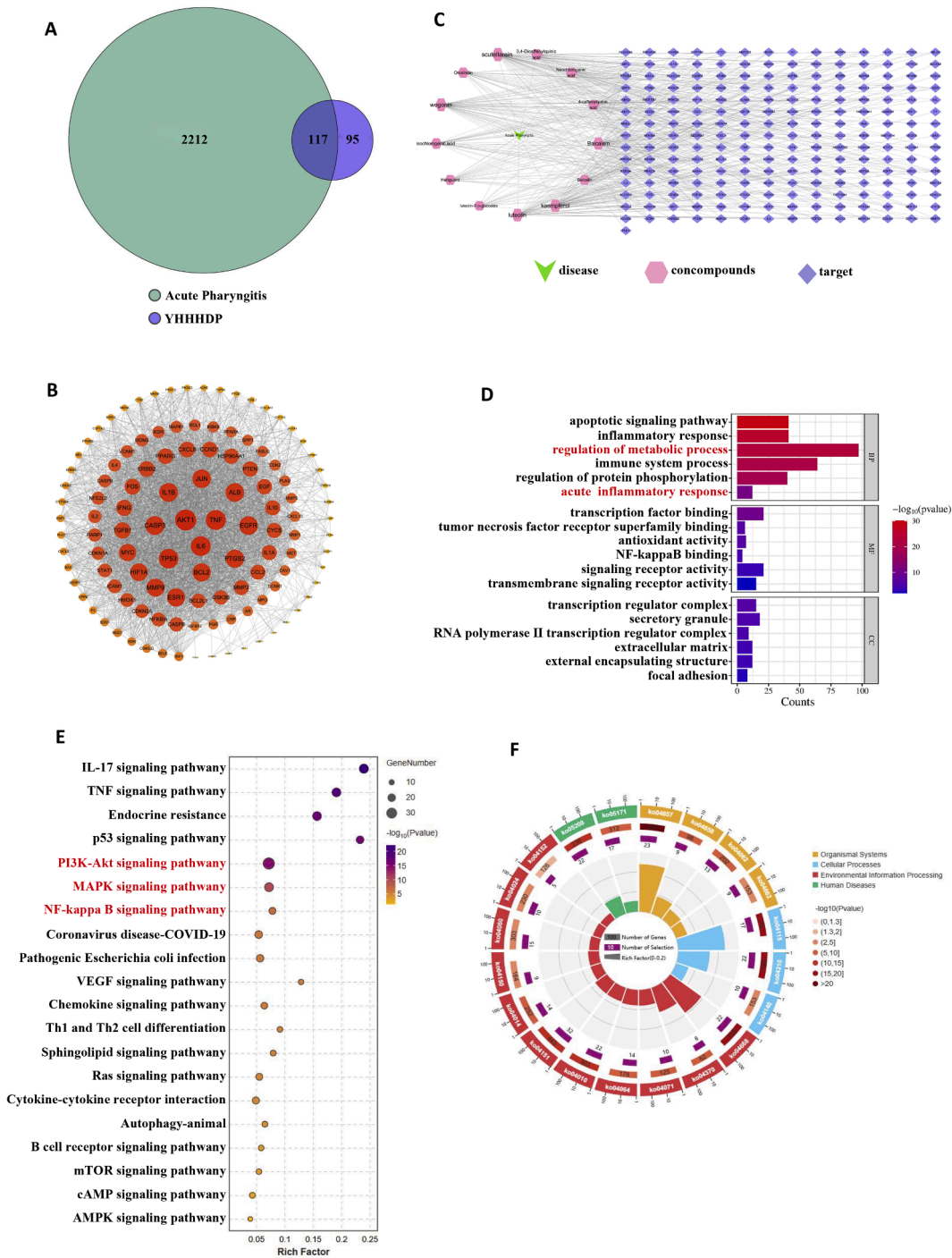


Fig. 2. Screening for Anti-Inflammatory Active Components from YHHHDP Using Network pharmacology Coupled with UHPLC-Q-Exactive Plus MS: (A) Venn diagram. (B) Protein-protein interaction network. (C) Disease-compounds-targets (DCT) network diagram. (D) GO functional enrichment analysis. (E) Bubble plot of the results of the KEGG significant enrichment analysis. (F) KEGG pathway enrichment circle diagram.

were identified for AP. Using a Venn diagram analysis, we identified 117 overlapping targets (Fig. 2A). We performed a comprehensive PPI network analysis for 117 potential targets to investigate the potential mechanisms by which YHHHDP treats AP. TNF, AKT1, IL-6, TP53, EGFR, and PTGS2 emerged as the key targets mediating the therapeutic effects of YHHHDP (Fig. 2B). Furthermore, we constructed a disease-component-target (DCT) network by connecting AP to 12 potential active components and therapeutic targets (Fig. 2C). GO enrichment analysis indicated significant enrichment in pathways linked to acute inflammatory response, regulation of metabolic processes, regulation of immune system, and NF- κ B signaling pathway. Based on these results, YHHHDP may exert its therapeutic effects by modulating inflammatory and metabolic responses (Fig. 2D). Furthermore, KEGG pathway enrichment analysis identified 85 pathways, with significant enrichment in those linked to inflammation and metabolism, such as PI3K-Akt, MAPK, NF- κ B, tryptophan metabolism, and arachidonic acid metabolism. (Fig. 2E-F). Our findings suggest that YHHHDP can provide therapeutic benefits in AP by modulating tryptophan and arachidonic acid metabolism in vivo, thereby affecting the PI3K-AKT/NF- κ B/MAPK signaling cascade.

Establishment of serum and tissue metabolomic profiles and metabolic biomarkers in AP rats

Chromatograms for controls in serum and tissue were captured in both ionization modes, positive and negative (Fig. S4A-S4D). The chromatograms exhibited well-defined peaks with good signal-to-noise ratios under the employed detection conditions. The mass spectrometry data from serum and tissue samples of all experimental rats were analyzed using multivariate statistical methods, such as PCA and OPLS-DA, using SIMCA 13.0 software.

PCA score plots of serum samples showed a distinct difference among the various groups (Fig. 3A), indicating a clear trend in metabolite profiles. PCA analysis of tissue samples demonstrated a complete separation of the YHHHDP group from the model group, with the YHHHDP group clustering closer to the control group (Fig. 3D). This suggests that YHHHDP effectively ameliorated the metabolic effects of AP, promoting the recovery to a healthier metabolic state in rats.

The metabolic differences between the groups were elucidated using OPLS-DA (Fig. S5A-B and Fig. S6A-B). The OPLS-DA score plots demonstrated distinct differences among the experimental groups. The model parameters (Table S8) showed R² and Q² values greater than 0.5, indicating good model fit and predictive ability. 200 permutations were conducted to validate the robustness of the model. The raw R² and Q² values on the right side were consistently higher than the permuted values on the left side across all groups, and these values were close to 1 (Fig. S5C-D and Fig. S6C-D). This indicates that the constructed model did not exhibit overfitting, demonstrating strong predictive ability, reliability, and suitability for subsequent differential metabolite screening.

Based on the OPLS-DA model, potential metabolic markers were initially identified using a VIP score threshold of ≥ 1 . As a final step, differential metabolites were selected by FC criteria ($FC > 1.2$ or $FC < 0.83$) and p-value thresholds of < 0.05 (Fig. 3B-C and 3E-F). The metabolomic analysis revealed substantial differences in metabolite levels among the experimental groups. Various experimental groups exhibited substantial differences in metabolite levels based on the metabolomic analysis. Differential metabolites were identified using online databases, such as METLIN, HMDB, PubChem, MetaboLights, and LIPIDMAPS. Serum metabolomics analysis revealed 27 significantly altered metabolites (Table S9), including 20 upregulated and 7 down-regulated metabolites. Similarly, tissue metabolomics analysis revealed 42 significantly altered metabolites, including 34 upregulated and 8 downregulated metabolites (Table S10). The findings imply that YHHHDP's therapeutic effects are due to its modulation of these metabolites.

Analysis of potential metabolic pathways in AP

The clustered heatmap analysis offers a clear visual depiction of the variation in metabolic marker levels across different experimental groups (Fig. 3G-H). YHHHDP treatment demonstrated a tendency to influence these metabolic markers in comparison to the model group, hinting that YHHHDP may have a therapeutic role in AP by altering endogenous metabolites.

MetaboAnalyst 6.0 was used to analyze metabolic pathways modulated by YHHHDP in AP, with significance regarded as a p-value less than 0.05 or an impact over 0.01. In the serum, YHHHDP was found to affect 14 metabolic pathways, which include the metabolism of tryptophan, phenylalanine, and tyrosine (Fig. 3I). Tissue metabolism analysis showed that 10 major pathways were affected, with notable pathways being arachidonic acid metabolism, sphingolipid metabolism, and purine metabolism (Fig. 3J).

A study comparing serum and tissue metabolomes revealed shared metabolic pathways, including pyrimidine, glycerophospholipid, arginine, proline, alanine, aspartate, glutamate, tryptophan, glycine, serine, threonine metabolism, and primary bile acid biosynthesis. Importantly, tissue-specific metabolism revealed distinct pathways, including arachidonic acid metabolism, sphingolipid metabolism, purine metabolism, and retinol metabolism. In contrast, serum metabolomics identified distinct pathways, including the metabolism of phenylalanine and tyrosine, the biosynthesis of phenylalanine, tyrosine, and tryptophan, glutathione metabolism, steroid hormone biosynthesis, and arginine biosynthesis. These findings suggest a potential correlation between metabolic fluctuations in tissue and serum, suggesting that tissue metabolic alterations can be reflected in serum.

Furthermore, the study cross-analyzed the metabolic pathways obtained from metabolomics with those predicted by network pharmacology, identifying 5 common metabolic pathways. After a comprehensive analysis of the P-values of these five pathways, tryptophan metabolism and arachidonic acid metabolism were confirmed as the key metabolic pathways involved in the treatment of AP by YHHHDP (Fig. S7).

Identification and validation of key targets

A component-metabolites-targets-pathway network was created to comprehensively assess the targets and anti-inflammatory mechanism of YHHHDP in AP through integrated analyses of network pharmacology and metabolomics (Fig. 4A). The network consisted of 12 YHHHDP components, 52 differentially metabolites, 39 potential targets, and 10 potential pathways. We aimed to identify key targets and signaling pathways potentially mediating the therapeutic effects of YHHHDP in AP. Based on degree values, AKT1, TNF- α , and PTGS2 were identified as key targets.

Western blotting was employed to experimentally validate the primary targets of YHHHDP both in vivo and in vitro (Fig. 4B-C). The expression of P-AKT, TNF- α , and COX-2 was significantly elevated in the AP model. Nonetheless, YHHHDP treatment notably decreased the expression of these targets. In RAW264.7 cells stimulated with LPS, YHHHDP led to significant reductions in TNF- α , COX-2, and P-AKT expression.

Inhibitory effect of YHHHDP active components on the activity of TNF- α , AKT1, and COX-2

The results of the effect of active ingredients on TNF- α activity revealed that compared with the TNF- α group, the active ingredients of YHHHDP, including wogonoside, galuteolin, luteolin, and isochlorogenic acid B, enhanced the survival rate of L929 cells to varying degrees in a dose-dependent manner. However, they are not as active as the positive control drug UCB-9260 (Fig. 5A).

Based on the results of the single-concentration inhibition

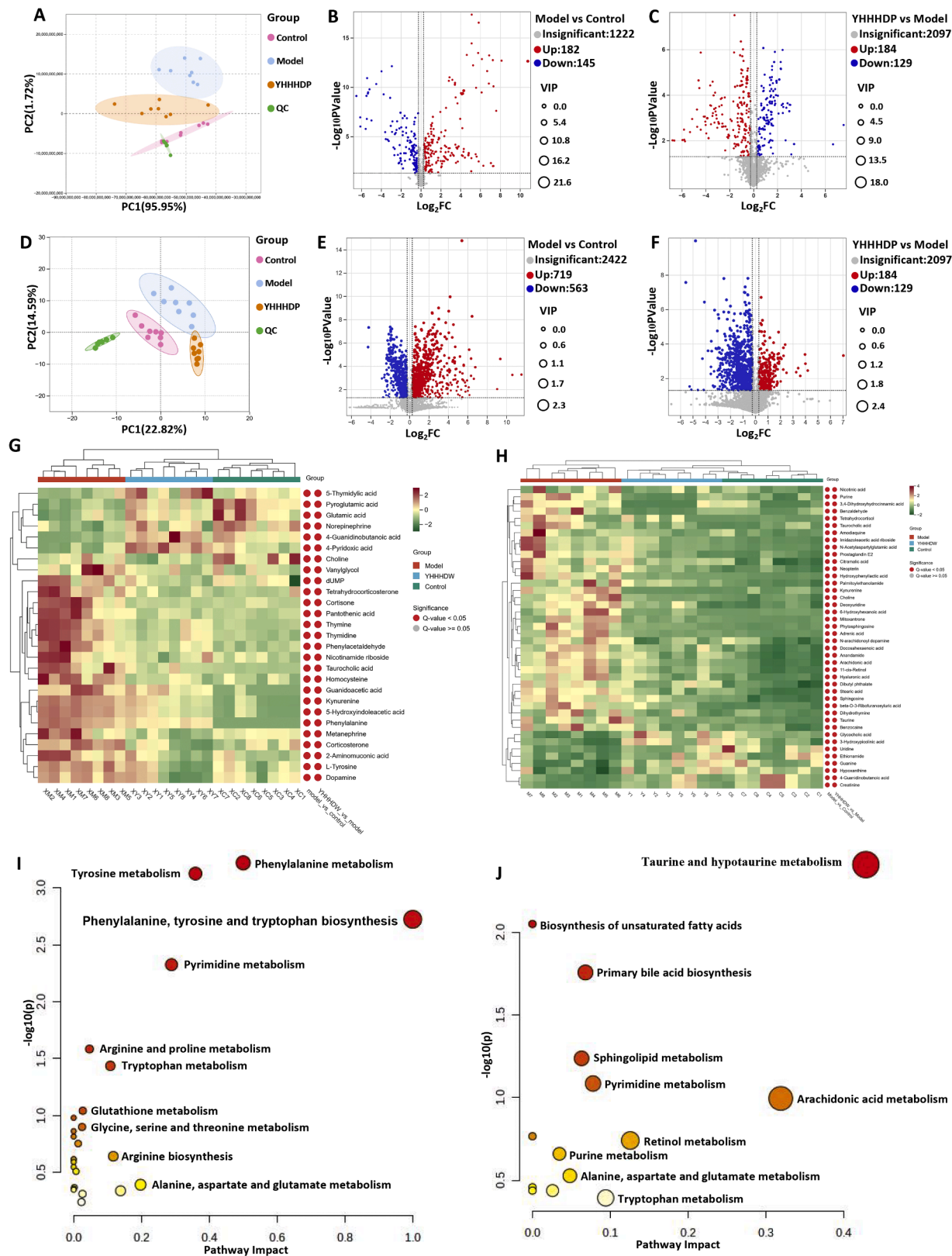


Fig. 3. Serum metabolomics and tissue metabolomics results. (A) PCA scores plot of Control, Model, YHHDP, and QC in serum. (B-C) Volcano plot in serum. (D) PCA scores plot of Control, Model, YHHDP, and QC in tissue. (E-F) Volcano plot in tissue. (G) Heatmaps of differential serum metabolites. (H) Heatmaps of differential tissue metabolites. (I) Metabolic pathway analysis of identified potential serum biomarkers. (J) Metabolic pathway analysis of identified potential tissue biomarkers.

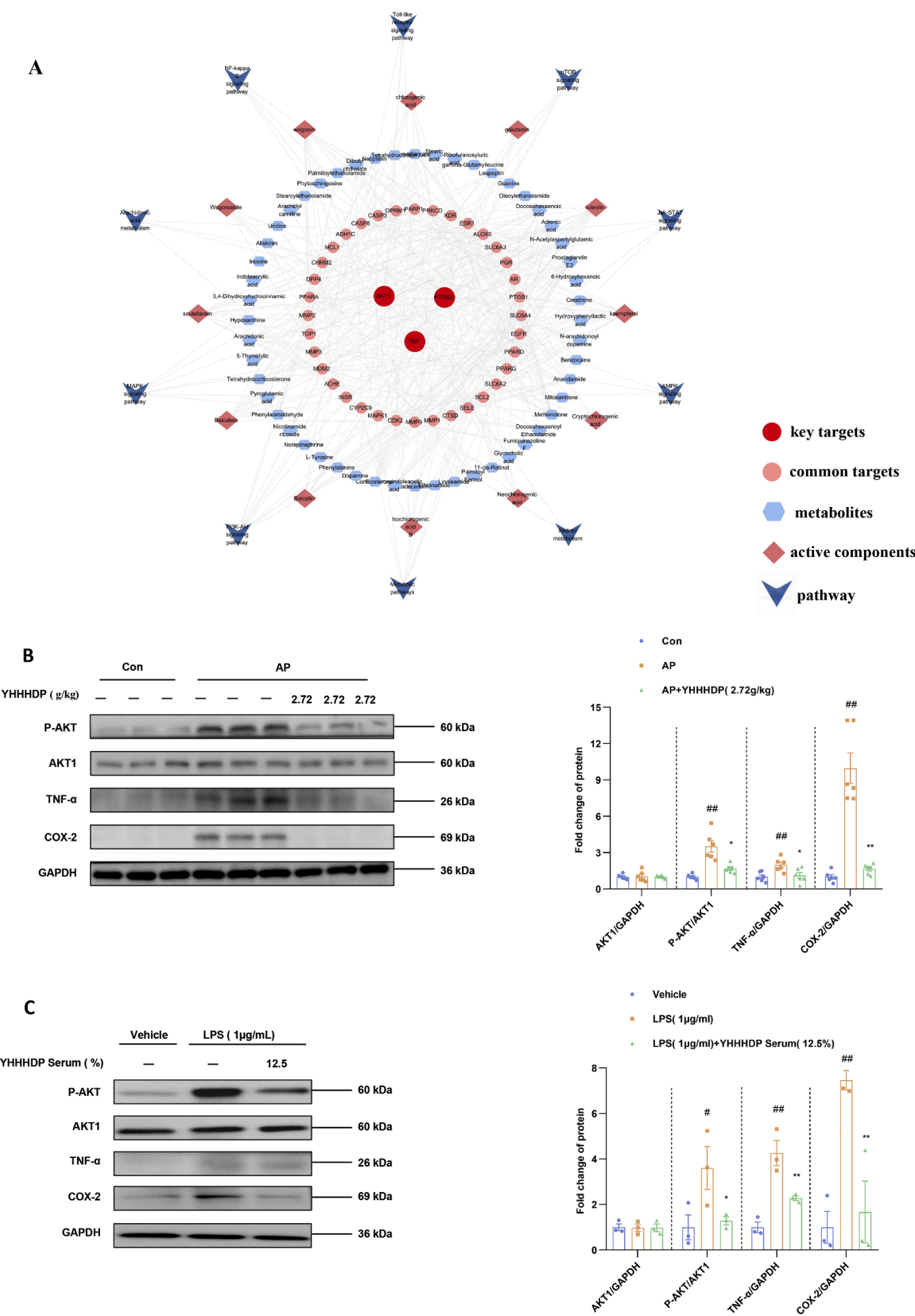


Fig. 4. Integrated Metabolomics and Network Pharmacology Analysis Identifies Key Targets of YHHHDP. (A) The active components-metabolites-targets-pathway association of with AP. (B) Western blot is used to evaluate the key targets AKT1, p-AKT, TNF- α and COX-2 levels in pharyngeal tissue ($n = 6$). (C) The protein expressions of RAW264.7 cell for AKT1, p-AKT, TNF- α ,and COX2($n = 3$) # $p < 0.05$, ## $p < 0.01$ vs. the control group; * $p < 0.05$, ** $p < 0.01$ vs. the model group.

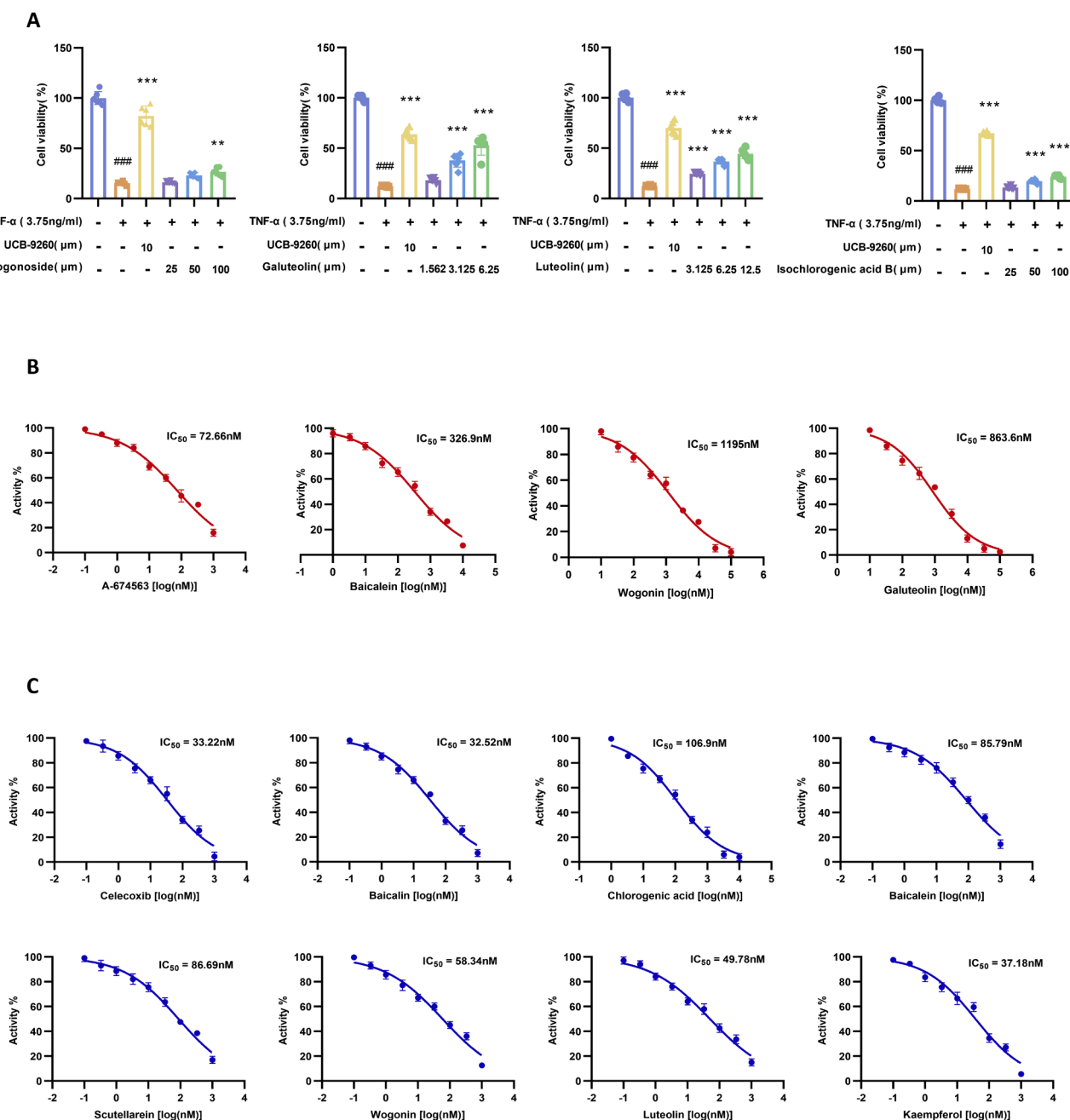


Fig. 5. YHHHDP active ingredients of functional assays. (A) Effects of four compounds on inhibition of TNF- α -mediated viability of L929 cells. (B) Dose-response curves of in vitro inhibition assays for A-67,453 and three AKT1-inhibitory compounds; (C) Dose-response curves of in vitro inhibition assays for celecoxib and seven COX-2-inhibitory compounds.

experiments, we selected active ingredients with an inhibition rate exceeding 50 % for further IC₅₀ experiments (Table S11 and Table S12).

The effects of the three active ingredients of YHHHDP on AKT1 activity, including their dose-response curves and IC₅₀ values, are illustrated in Fig. 5B. The IC₅₀ values of baicalein, wogonin and galuteolin were 326.9 nm, 1195 nm and 863.6 nm, respectively, and A-674,563 was used as a positive control with an IC₅₀ value of 72.66 nm.

Fig. 5C shows the dose-response curves and IC₅₀ values of the seven active ingredients of YHHHDP in relation to COX-2 activity. Celecoxib was used as positive control with an IC₅₀ value of 33.22 nm. The IC₅₀ values of baicalin, chlorogenic acid, baicalein, scutellarein, wogonin, luteolin, and kaempferol were 32.52 nm, 106.9 nm, 85.79 nm, 86.69 nm, 58.34 nm, 49.78 nm, and 37.18 nm, respectively.

The active ingredients interacting with TNF- α , AKT1, and COX-2

were verified based on Molecular docking, BLI, and CETSA

First, we performed molecular docking of the components that exhibited inhibitory effects on TNF- α , AKT1, and COX-2 activities, respectively. A more negative binding energy indicates a stronger interaction between the active ingredient and the target (Krishnasamy, 2024). Generally, binding affinity is deemed good when it is -5.0 kcal/mol or lower (Francis et al., 2024; Qian et al., 2020). The molecular docking results are shown in Fig. 6A, 7A, 8A and 9A, where wogonoside, galuteolin, luteolin, and isochlorogenic acid B displayed the strongest binding to TNF- α ; galuteolin, baicalein, and wogonin showed the strongest binding to AKT1; and baicalin, chlorogenic acid, baicalein,

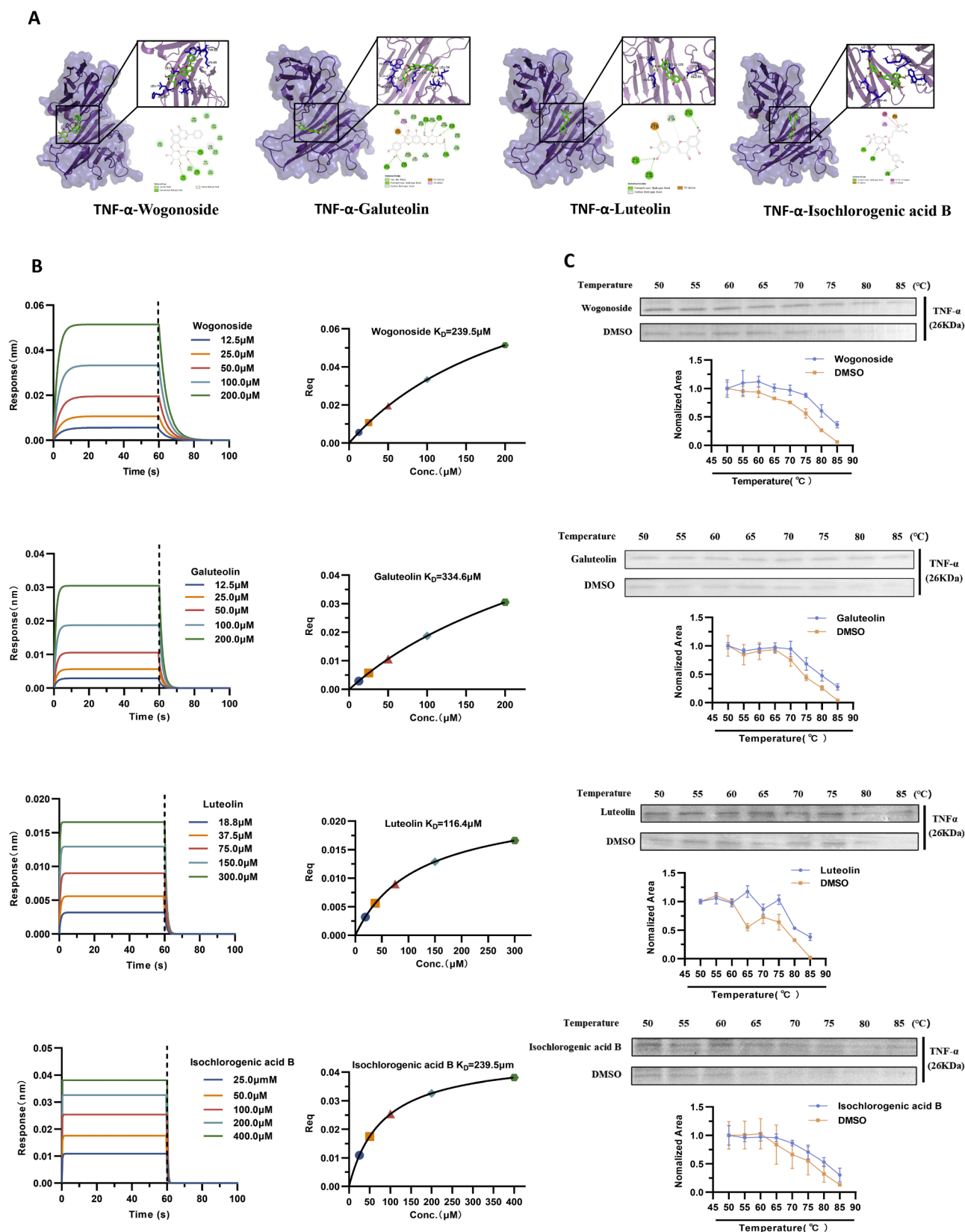


Fig. 6. The interactions between wogonoside, galuteolin, luteolin, and isochlorogenic acid B with TNF- α were verified using molecular docking, BLI, and CETSA. (A) Molecular docking of wogonoside, galuteolin, luteolin, and isochlorogenic acid B with TNF- α . (B) Realtime kinetic binding parameters of wogonoside, galuteolin, luteolin, and isochlorogenic acid B interacting with TNF- α , as determined by BLI. (C) CETSA was performed at temperatures ranging from 50 °C to 85 °C to examine the thermal stability of the interactions between TNF- α and wogonoside, galuteolin, luteolin, and isochlorogenic acid B ($n = 3$).

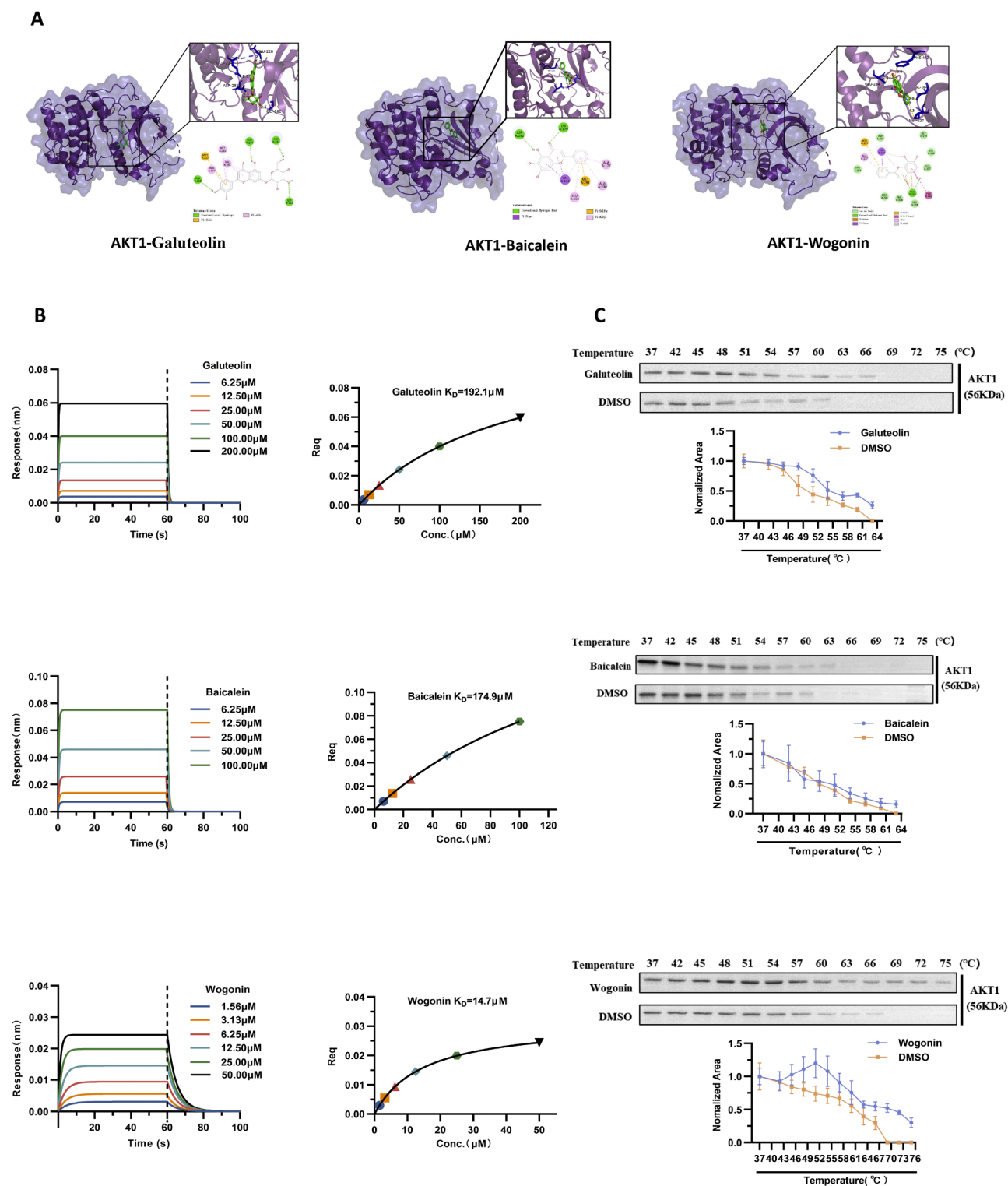


Fig. 7. The interactions between galuteolin, baicalein, and wogonin with AKT1 were verified using molecular docking, BLI, and CETSA. (A) Molecular docking of galuteolin, baicalein, and wogonin with AKT1. (B) Realtime kinetic binding parameters of galuteolin, baicalein, and wogonin interacting with AKT1, as determined by BLI. (C) CETSA was performed at temperatures ranging from 37 °C to 75 °C to examine the thermal stability of the interactions between AKT1 and galuteolin, baicalein, and wogonin ($n = 3$).

scutellarein, wogonin, luteolin, and kaempferol showed the strongest binding to COX-2. Additionally, we calculated the mean and standard deviation of the three molecular docking experiments, with standard deviations ranging from 0.08 to 0.42 kcal/mol. We also determined the predicted binding values (pKi in μM) (Li et al., 2024a). The mean, standard deviation, and predicted binding values of the three molecular

docking experiments are summarised in Table S13.

Next, we used BLI and CETSA to further validate the molecular docking results.

Analysis of the BLI results, as shown in Fig.6B, Fig.7B, Fig.8B and Fig.9B, indicated that the optical thickness response increased with the drug dose. The dissociation constants (Kd) were determined for each

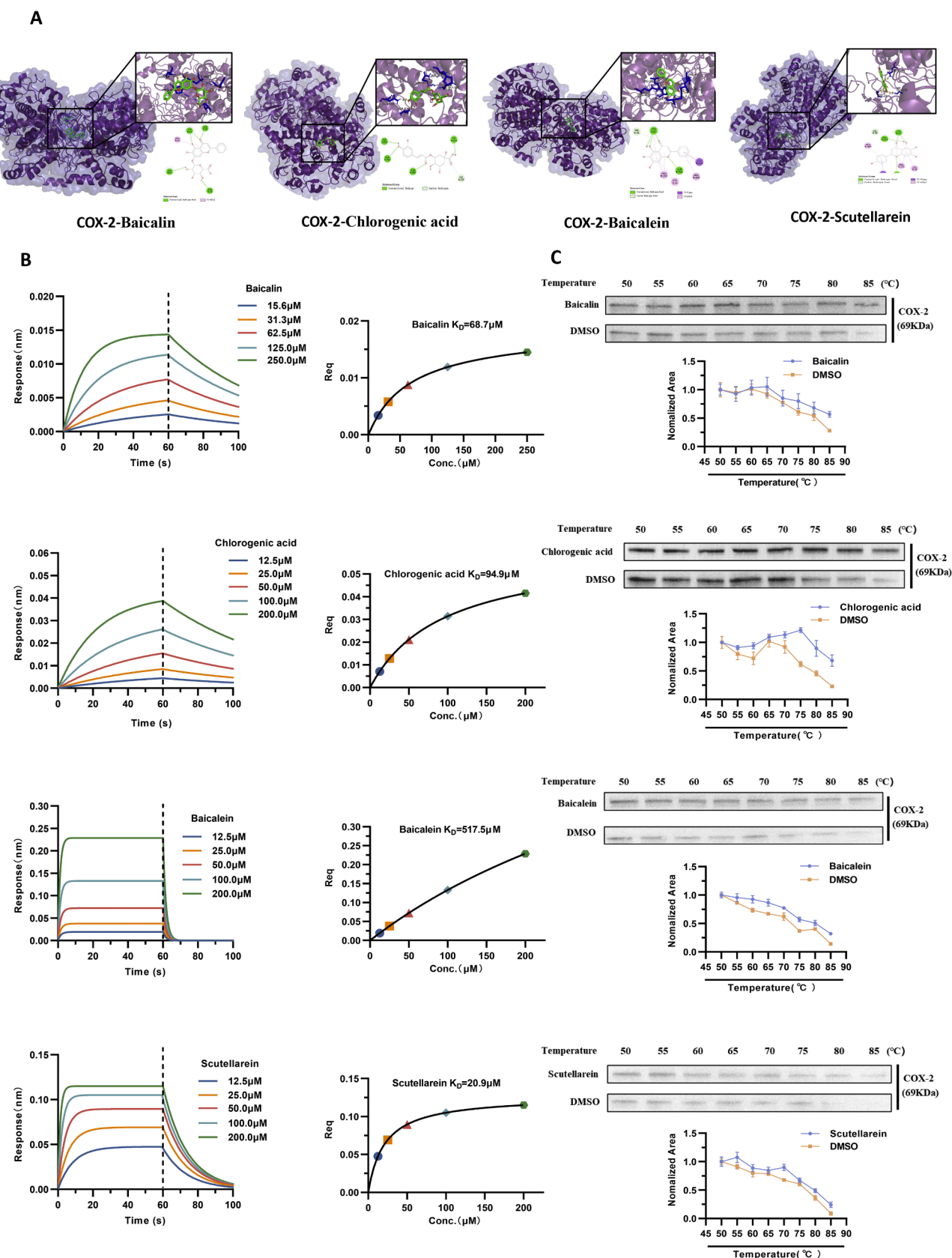


Fig. 8. The interactions between baicalin, chlorogenic acid, baicalein, and scutellarein with COX-2 were verified using molecular docking, BLI, and CETSA. (A) Molecular docking of baicalin, chlorogenic acid, baicalein, and scutellarein with COX-2. (B) Realtime kinetic binding parameters of baicalin, chlorogenic acid, baicalein, and scutellarein interacting with COX-2, as determined by BLI. (C) CETSA was performed at temperatures ranging from 50 °C to 85 °C to examine the thermal stability of the interactions between COX-2 and baicalin, chlorogenic acid, baicalein, and scutellarein ($n = 3$).

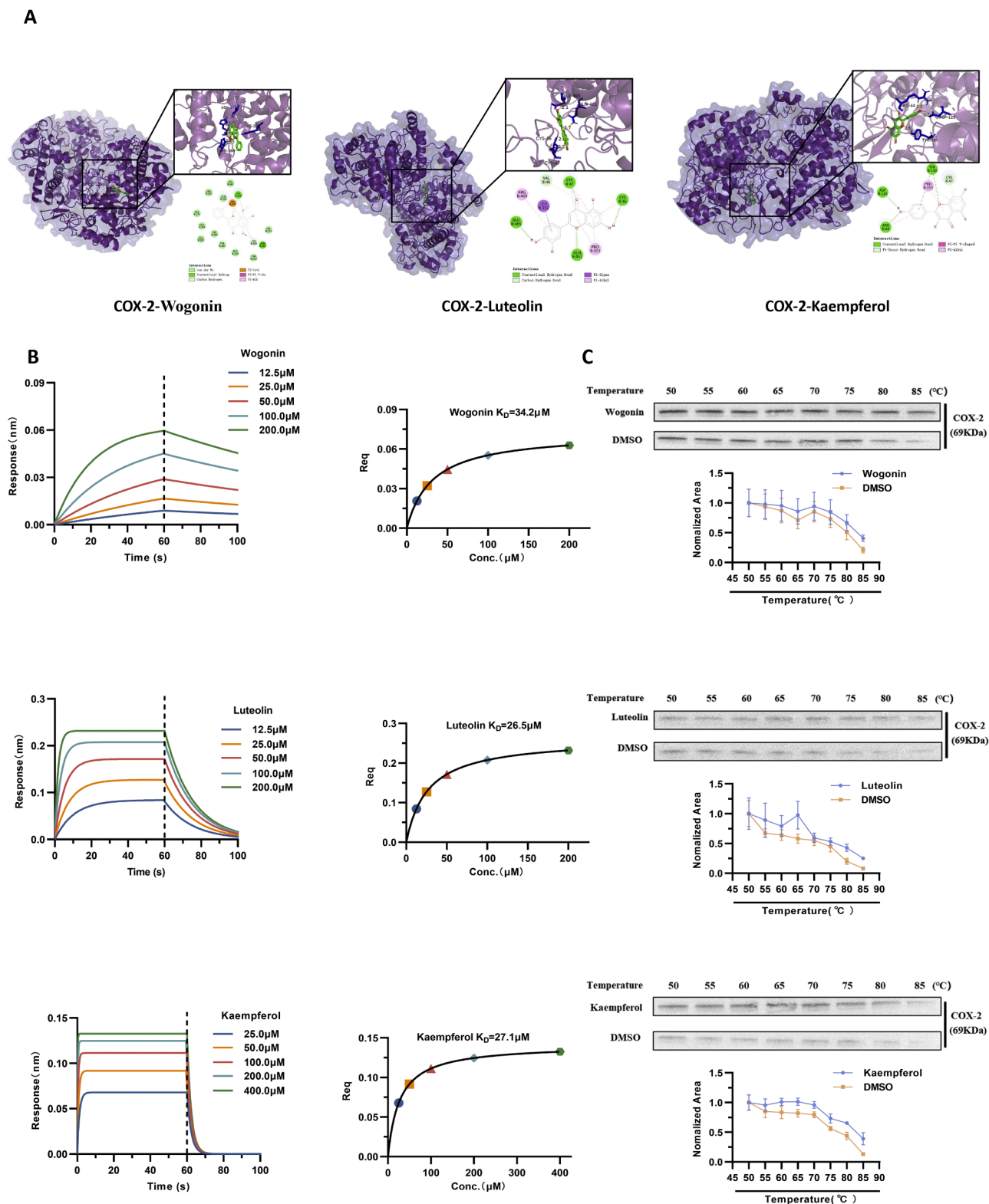


Fig. 9. The interactions between wogonin, luteolin, and Kaempferol with COX-2 were verified using molecular docking, BLI, and CETSA. (A) Molecular docking of wogonin, luteolin, and Kaempferol with COX-2. (B) Realtime kinetic binding parameters of wogonin, luteolin, and Kaempferol interacting with COX-2, as determined by BLI. (C) CETSA was performed at temperatures ranging from 50 °C to 85 °C to examine the thermal stability of the interactions between COX-2 and wogonin, luteolin, and Kaempferol ($n = 3$).

active ingredient in direct interaction with the target.

According to biophysical principles, proteins show thermal stabilization upon ligand binding; thus, we performed CETSA to assess if YHHHDP's active components interacted with TNF- α , AKT1, and COX-2.

The compounds wogonoside, galuteolin, luteolin, and isochlorogenic acid B exhibited higher thermal stability of TNF- α compared to the DMSO group in the temperature range of 50 °C to 85 °C (Fig. 6C). Galuteolin, baicalein, and wogonin exhibited higher thermal stability of

AKT1 compared to the DMSO group in the temperature range of 37 °C to 75 °C(Fig.7C). Baicalin, chlorogenic acid, baicalein, scutellarein, wogonin, luteolin, and kaempferol exhibited higher thermal stability of COX-2 compared to the DMSO group in the temperature range of 50 °C

to 85 °C(Fig.8C and Fig.9C).

These in vitro validation results indicate direct and reversible interactions between the active ingredients of YHHHDP and TNF- α , AKT1, and COX2. The findings also show that the same active ingredient can

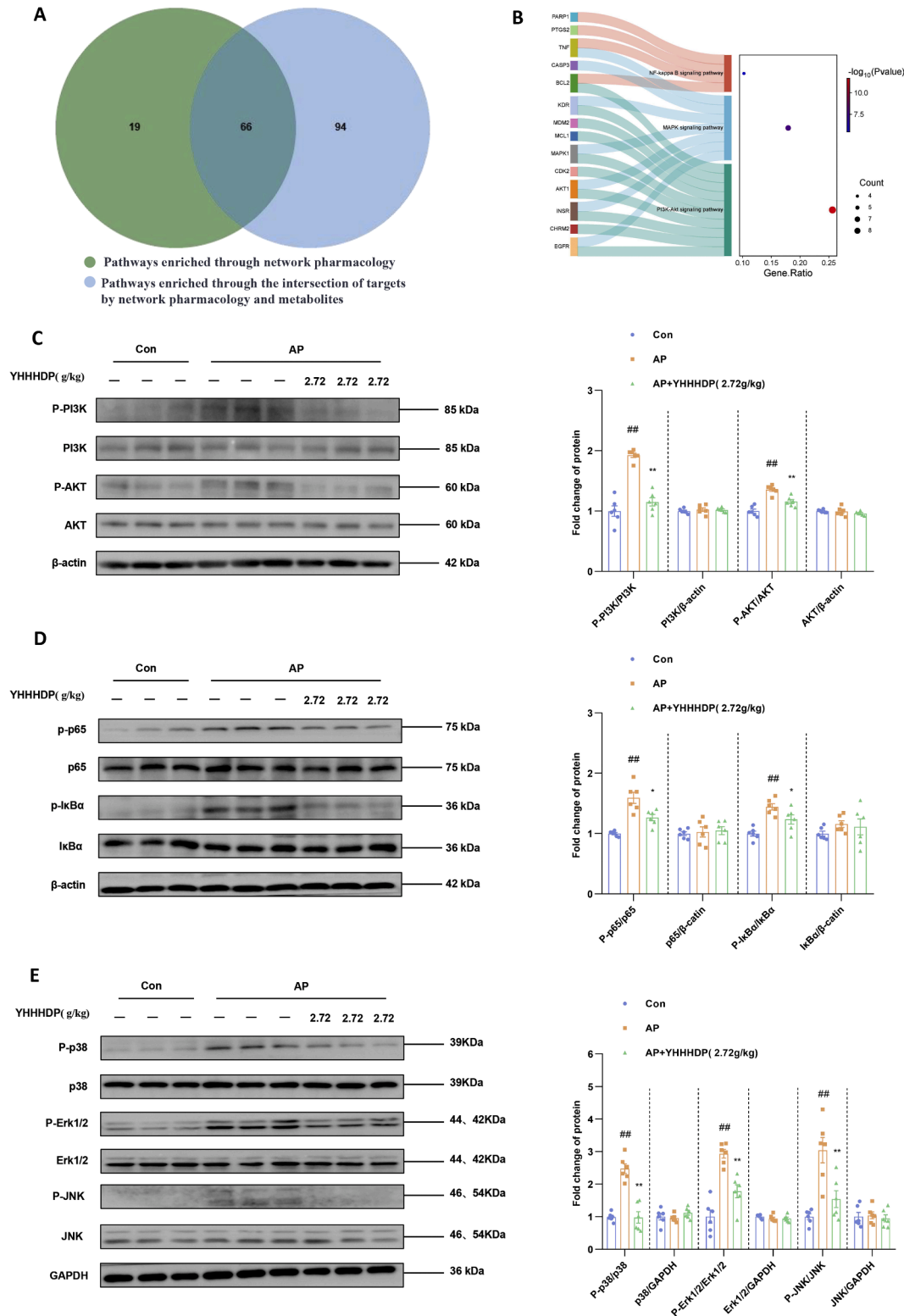


Fig. 10. YHHHDP suppressed the inflammatory response through blocking the PI3K-AKT/NF- κ B/MAPK pathway. (A) Venn diagram. (B) sankey dot plot. (C) Western blot is used to evaluate the effects of YHHHDP on the PI3K-AKT signaling pathway in pharyngeal tissue of AP model rats ($n = 6$). (D) The effects of YHHHDP on the NF- κ B signaling pathway in pharyngeal tissue ($n = 6$). (E) The effects of YHHHDP on the MAPK signaling pathway in pharyngeal tissue ($n = 6$). $\#p < 0.05$, $\#\#p < 0.01$ vs. the control group; $*p < 0.05$, $**p < 0.01$ vs. the model group.

interact with multiple targets, further demonstrating the multicomponent and multitarget nature of herbal medicines and the potential for synergistic interactions among the herbal ingredients.

In vivo and in vitro validation of the mechanism of action of YHHHDP

Through the integration of network pharmacology and metabolomics analysis, a network connecting ingredients, metabolites, targets, and pathways was established. The 39 potential targets identified in the ingredient-metabolite-target-pathway network were analyzed through KEGG enrichment, revealing 160 signaling pathways. Performing intersection analysis with the pathways predicted by network

pharmacology narrowed down the focus to 66 relevant pathways (Fig. 10A). Based on P-value and an inflammatory perspective, we identified the PI3K-AKT, NF- κ B, and MAPK pathways as the key pathways modulated by YHHHDP to reduce inflammation (Fig. 10B).

Subsequently, we verified the key pathways using Western blotting. Compared to control rats, model rats showed significantly higher phosphorylation levels of PI3K, AKT, p65, I κ B α , JNK, p38, and Erk1/2 in pharyngeal tissues, with no significant change in total protein levels (Fig. 10C-E). This suggests that activation of the PI3K/AKT/NF- κ B/MAPK pathway is associated with severe inflammation. YHHHDP treatment notably decreased the phosphorylation of these proteins, suggesting its ability to reduce inflammation by blocking the PI3K/AKT/NF- κ B/MAPK signaling pathway.

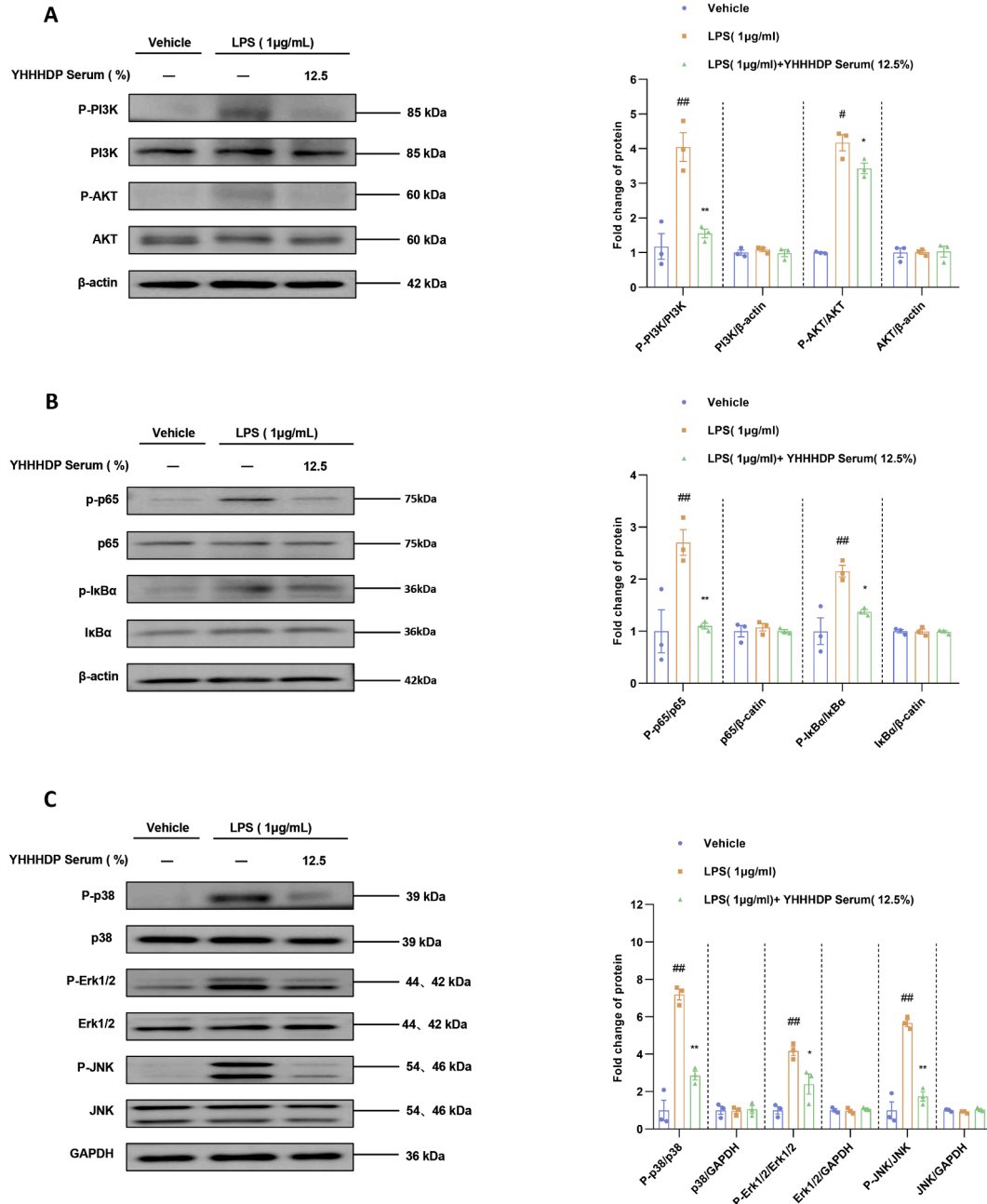


Fig. 11. YHHHDP serum suppressed the inflammatory response through blocking the PI3K-AKT/NF- κ B/MAPK pathway. (A) Western blot is used to evaluate the effects of YHHHDP serum on the PI3K-AKT signaling pathway in RAW264.7 cell ($n = 3$). (B) The effects of YHHHDP on the NF- κ B signaling pathway in RAW264.7 cell ($n = 3$). (C) The effects of YHHHDP on the MAPK signaling pathway in RAW264.7 cell ($n = 3$). # $p < 0.05$, ## $p < 0.01$ vs. the Vehicle group; * $p < 0.05$, ** $p < 0.01$ vs. the LPS group.

The phosphorylation of PI3K, AKT, p65, I κ B α , JNK, p38, and Erk1/2 was significantly enhanced by LPS in RAW264.7 cells, but the overall protein expression did not change. The administration of YHHHDP markedly decreased the phosphorylation levels of the aforementioned proteins (Fig. 11A–C). YHHHDP suppresses inflammation by inhibiting the PI3K-AKT/NF- κ B/MAPK pathway in RAW264.7 cells.

Discussion

AP is an acute, non-specific inflammation of the pharyngeal mucosa, primarily caused by viral and bacterial infections and chemical irritation. Ammonia irritation can rapidly induce local vasodilation and congestion, compromising the integrity of pharyngeal tissues and inducing a massive infiltration of inflammatory cells (Coutinho et al., 2021; Miao et al., 2019). Many studies have reported a significant association between AP and metabolic disorders in vivo. YHHHDP, a TCM formulation known for its heat-clearing and detoxifying effects, is primarily used to treat AP and various upper respiratory tract infections, showing significant clinical efficacy. However, the active components and mechanisms by which YHHHDP treats AP remain unclear. In addition, it has not been reported whether YHHHDP can treat AP by regulating metabolic disorders.

The development of pharyngitis is closely associated with inflammation. TNF- α is pivotal in acute inflammatory processes, as it activates neutrophils and lymphocytes, enhances vascular permeability, and modulates the overall inflammatory response (Song et al., 2018). Furthermore, IL-1 β induces the production of additional cytokines and chemokines, thus facilitating immune cell infiltration (Melton and Qiu, 2020). IL-6 and PGE2 possess both pro-inflammatory and anti-inflammatory effects; however, they primarily exert pro-inflammatory effects in AP (Wang et al., 2024). The findings of previous studies were consistent with our findings, demonstrating that in a rat model of AP, pharyngeal tissues exhibit severe pathological changes, characterized by mucosal epithelial cell desquamation, sub-mucosal glandular hyperplasia, inflammatory cell infiltration, and congestion. Additionally, treatment with YHHHDP markedly reduced pharyngeal histopathological damage, accompanied by a significant decrease in the levels of pro-inflammatory cytokines. The combined findings reveal the significant anti-inflammatory impact of YHHHDP in treating AP.

Macrophages, as key mediators of inflammation, are essential for the initiation, maintenance, and resolution of inflammatory responses (Chen et al., 2023; Schett and Neurath, 2018). Therefore, RAW264.7 cells are commonly used as a cell model for AP to investigate the anti-inflammatory effect (Hong et al., 2025; Huang et al., 2024). After being stimulated with LPS, macrophages release high levels of pro-inflammatory cytokines, including TNF- α and IL-1 β . The excessive production of cytokines stimulates the expression of the *iNOS* genes in macrophages, which induces NO production (Peng et al., 2019). In RAW264.7 cells, LPS induction increased TNF- α , IL-1 β , IL-6, and NO levels, aligning with earlier research. YHHHDP significantly reduced these inflammatory markers and alleviated the inflammatory response.

This study's findings showed that baicalin, baicalein, scutellarein, wogonoside, wogonin, galuteolin, chlorogenic acid, luteolin, kaempferol, and isochlorogenic acid B are the primary active components of YHHHDP used in treating AP. This aligns with previous studies, reporting that these components exhibit good anti-inflammatory activity when used separately. We also explored the key targets of these active ingredients, including TNF- α , AKT1, and COX-2. As a key cytokine bridging inflammation, TNF is crucial in both inflammatory and immune responses (Bekusova et al., 2022). It activates inflammatory cells and induces their proliferation and differentiation, regulating apoptosis and physiological processes. However, high levels of TNF can lead to tissue damage and exacerbate inflammation. Our study revealed that YHHHDP can significantly lower the serum levels of TNF- α and suppress the tissue expression of TNF- α , thereby effectively alleviating

inflammation. In vitro validation experiments also demonstrated that chlorogenic acid, lignoceroside, and baicalein interact with TNF- α and inhibit its activity.

AKT1, alternatively referred to as protein kinase B (PKB), is a serine/threonine kinase involved in the PI3K-AKT signaling cascade. Phosphorylation activates AKT, and its activation is strongly linked to the inflammatory response. Notably, AKT1 has been shown to promote inflammatory responses (Abdalla et al., 2015; Huang et al., 2022). The present study indicated that the active ingredients of YHHHDP interact with and inhibit AKT1, downregulate AKT phosphorylation, and attenuate the inflammatory response.

The COX-2 gene encodes a protein critical for regulating inflammatory responses. This gene is crucial for inflammation and is considered a potential target for treating various inflammatory diseases (Gobalan et al., 2023; Vogel et al., 2014). COX-2 expression is typically low in normal conditions and increases rapidly in response to infection, injury, or stress. Studies have shown that reducing COX-2 expression can substantially suppress the inflammatory response (Astakhova et al., 2019). These findings confirm our observation that the active ingredients of YHHHDP can effectively alleviate inflammation by interacting with COX-2, inhibiting its enzymatic activity, and downregulating COX-2 levels.

The identification of active ingredient targets in TCM is crucial for elucidating their pharmacodynamic mechanisms. The emergence of biophysics, chemical biology, and other advanced technologies has provided robust methods for identifying the direct targets of active ingredients in TCM. In this study, functional assays, such as enzymatic inhibition, and binding studies, including molecular docking, BLI, and CETSA, were employed to validate the targets of active ingredient action from multiple perspectives. Molecular docking, BLI, and CETSA are currently widely used to investigate the interactions between active ingredients (small molecule natural drugs) and their targets in TCM (Kalyanaraman et al., 2024; Tu et al., 2023; Zhang et al., 2024).

Subsequently, we comprehensively analyzed the mechanism of action of YHHHDP in treating AP using a combination of network pharmacology and metabolomics. The findings indicated that YHHHDP can modulate AP via various pathways, especially those related to inflammation like PI3K-AKT, NF- κ B, and MAPK, as well as pathways related to metabolism such as tryptophan and arachidonic acid metabolism.

PI3K-AKT, NF- κ B, and MAPK pathways are crucial for controlling inflammation and play a major role in the development and progression of AP. They trigger a rapid inflammatory reaction, resulting in the release of pro-inflammatory cytokines (Jia et al., 2020; Wang et al., 2024).

The PI3K-AKT pathway, interacting with NF- κ B, is crucial for regulating inflammation (Qiu et al., 2020). The encoding of pro-inflammatory cytokines relies heavily on transcription factors such as NF- κ B. Upon activation, NF- κ B undergoes phosphorylation and the I κ B protein undergoes proteasomal degradation, which releases and phosphorylates NF- κ B dimers. These dimers then translocate from the cytoplasm to the nucleus, where they modulate the expression of pro-inflammatory cytokines, such as TNF- α and IL-1 β , thereby initiating an inflammatory response (Peng et al., 2019). The research demonstrated that YHHHDP markedly reduces the phosphorylation of PI3K, AKT, I κ B α , and p65 in the pharyngeal tissues of rats with ammonia-induced AP, indicating its potential to inhibit proteins in the PI3K-AKT/NF- κ B pathway and reduce inflammation. Additionally, the activation of the MAPK signaling pathway, which includes the Erk1/2, JNK, and p38 subfamilies, can induce the production of pro-inflammatory factors. Activation of this pathway is associated with the phosphorylation of p38, Erk1/2, and JNK (Li et al., 2023; Zhou et al., 2022). In this study, a significant increase in phosphorylation levels of JNK, Erk1/2, and p38 was observed in the pharyngeal tissues of AP rats. Treatment with YHHHDP effectively inhibited the phosphorylation of these MAPK components. These results indicated that YHHHDP can attenuate inflammatory responses by blocking the MAPK signaling

pathway.

Although previous studies have established a strong correlation between the inflammatory response in AP and endogenous metabolic alterations *in vivo*, the precise mechanism underlying this association remains elusive (Ding et al., 2023). Earlier studies have shown that the primary metabolic pathways involved in the pathogenesis of AP include amino acid metabolism, lipid metabolism, and nucleotide metabolism, which aligns with the results of the current study (Ding et al., 2023). Our findings showed that treating AP with YHHHDP was significantly associated with tryptophan metabolism and arachidonic acid metabolism.

Tryptophan metabolism is intricately linked to the inflammatory response, proceeding through three main pathways: (1) tryptophan degradation, (2) the kynurenine pathway, and (3) the 5-hydroxytryptamine (serotonin) production pathway. Severe inflammatory response is often associated with increased amino acid metabolism (Angus et al., 2022; Hu et al., 2020). Consistent with previous studies, the AP group displayed significantly elevated levels of 5-hydroxyindoleacetic acid, kynurenine, and 2-aminohexadienoic acid, key metabolites of tryptophan metabolism. Target prediction of tryptophan metabolites identified COX-2 as a key target. Tryptophan metabolites modulate the NF- κ B and MAPK pathways by activating AhR, which reduces inflammation by modulating NF- κ B and MAPK phosphorylation (Jiang et al., 2023). This aligns with our findings showing that YHHHDP can effectively regulate the levels of tryptophan metabolites, ameliorate tryptophan metabolism disorders, and significantly reduce the phosphorylation of NF- κ B and MAPK, finally mitigating inflammation.

Arachidonic acid metabolism is a major metabolic pathway. Arachidonic acid serves as a pro-inflammatory precursor implicated in various metabolic disorders (Sztolsztener et al., 2020). Arachidonic acid triggers

intracellular signaling pathways by interacting with receptors on inflammatory cells, leading to increased production and release of pro-inflammatory cytokines, thereby worsening the inflammatory response (Trostchansky and Rubbo, 2017). Key enzymes of arachidonic acid metabolism, such as COX-2, 5-lipoxygenase (5-LOX), and cytochrome P450 (CYP) 4A, play crucial roles in human inflammatory diseases. Previous studies have demonstrated that COX-2/PGE2 and 5-LOX/LTB4 activate the PI3K-AKT signaling pathway, an upstream activator of NF- κ B, via their respective receptors EP2 and BLT1, enhancing p65 and STAT1 phosphorylation and promoting inflammation (Sztolsztener et al., 2020; Zheng et al., 2014). This is consistent with the findings of a present study, which demonstrated that the active ingredient of YHHHDP can interact with COX-2 and significantly reduce the levels of PGE2 in pharyngeal tissues. The downregulation of PGE2 further reduced the expression of PI3K, AKT, and NF- κ B, thereby alleviating the inflammatory response.

Numerous studies have reported the use of TCM in the treatment of AP. In this study, the active components, key targets, and mechanism of action of YHHHDP in the treatment of AP were investigated for the first time from the perspectives of metabolic disorders and inflammatory responses and our findings were validated *in vitro*. Specifically, the active ingredients of YHHHDP were shown to regulate tryptophan and arachidonic acid metabolism and inhibit the PI3K-AKT/NF- κ B/MAPK signaling pathway by interacting with key targets, such as TNF- α , AKT1, and COX-2 (Fig. 12). Thus, they effectively ameliorate the symptoms of AP. Meanwhile, this study demonstrated that TCM synergistically affects multiple components, targets, and pathways. A single component can act on multiple targets and regulate several pathways to exert therapeutic effects. This suggests that we can further explore components with multi-target and multi-pathway effects from Chinese medicine and

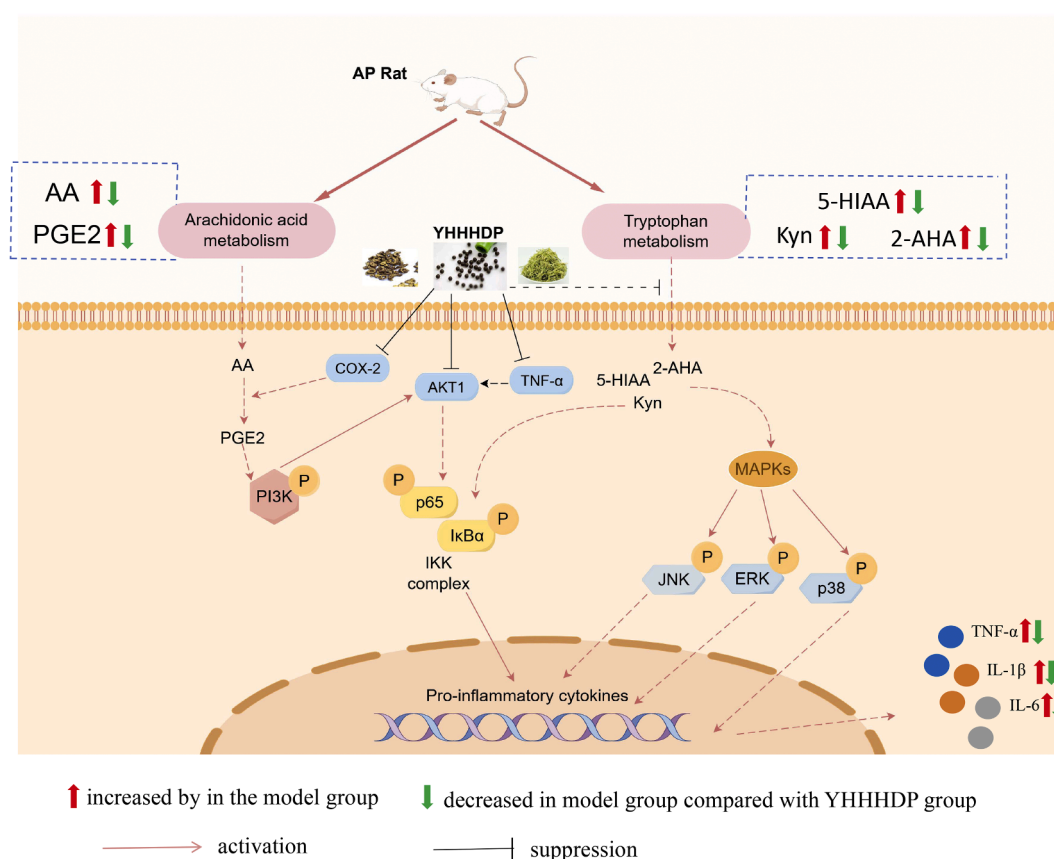


Fig. 12. The key mechanism of the anti-inflammatory effect of YHHHDP is that its active ingredients regulate tryptophan and arachidonic acid metabolism by inhibiting key targets TNF- α , AKT1, and COX2. This inhibition also suppresses the activation of the PI3K-AKT/NF- κ B/MAPK signalling pathway, thereby reducing the release of inflammatory factors.

develop innovative drugs.

Our study has initially uncovered the active components of YHHHDP and their mechanisms for treating acute pharyngitis, but there are still some limitations in the current research. For example, we have not used targeted metabolomics techniques to validate the metabolomics results. Additionally, for the mechanism studies, we have not employed relevant agonists or inhibitors for more in-depth exploration. It is well known that TCM works through multiple pathways. This study focuses primarily on the PI3K-AKT/NF- κ B/MAPK signalling pathway, but other pathways may also contribute to the anti-inflammatory effects of YHHHDP. Therefore, in our follow studies, we will continue to further explore other potential signalling pathways or mechanisms underlying the anti-inflammatory effects of YHHHDP. Secondly, in this study, an animal model of ammonia induced AP was employed. Although the pathological damage and changes in serum cytokine levels in this model were somewhat similar to the clinical manifestations of AP, most clinical cases are caused by viruses or bacteria. Therefore, our followup study will further investigate models of AP induced by viral or bacterial agents. Finally, the lack of highquality prospective clinical trial studies presents numerous challenges in translating the efficacy of YHHHDP into clinical practice, particularly in terms of dosage standardization and safety in human populations (Shao et al., 2024). Consequently, in our subsequent indepth studies, we will focus more on the dosage standardization and safety in human populations of YHHHDP, including the development of new dosage forms and the exploration of additional indications (Mohanta et al., 2024). We will strive to obtain new clinical evidence and uncover new clinical value. Looking forward, we hope to provide a more solid and clear experimental basis for the clinical application of YHHHDP.

Conclusion

In this study, the therapeutic effect of YHHHDP was confirmed in both a rat model of acute pharyngitis and a RAW264.7 cell model. The mechanism of action was elucidated, and potential active ingredients were predicted using network pharmacology and metabolomics. YHHHDP inhibits inflammation and improves the symptoms of acute pharyngitis by inhibiting TNF- α , AKT1, and COX-2, regulating the metabolism of arachidonic acid and tryptophan, and inhibiting the activation of the PI3K-AKT/NF- κ B/MAPK signaling pathway. The active ingredients of YHHHDP for the treatment of acute pharyngitis include baicalin, baicalein, scutellarein, wogonoside, wogonin, galuteolin, chlorogenic acid, luteolin, kaempferol, and isochlorogenic acid B.

Abbreviations

AP, Acute pharyngitis; AKT, protein kinase B; COX-2, Cyclooxygenase-2; Dex, Dexamethasone; Erk, Extracellular signal-regulated kinase; ELISA, Enzyme-linked immunosorbent assay. TNF- α , Tumor necrosis factor- α ; GO, gene ontology; IL-1 β , Interleukin-1 β ; IL-6, Interleukin-6; I κ B α , inhibitor of nuclear factor kappa-B; JNK, c-Jun N-terminal kinases; KEGG, Kyoto Encyclopedia of Genes and Genomes; LPS, Lipopolysaccharide; MAPK, mitogen activated protein kinase; NO, Nitric oxide; NF- κ B, nuclear factor kappa-B; OPLS-DA, Orthogonal partial least squares discriminant analysis; PGE2, Prostaglandin E2; PCA, Principal components analysis; PI3K, phosphoinositide 3-kinase; QCs, Quality control samples; SD, Sprague Dawley; TCM, Traditional Chinese medicine; VIP, Variable importance for projection; YHHHDP, Yinhuang Hanhua Dropping Pills;

All data were generated in-house, and no paper mill was used. All authors agree to be accountable for all aspects of work ensuring integrity and accuracy.

Author agreement

We the undersigned declare that this manuscript entitled "Yinhuang

Hanhua Dropping Pills's effect on Acute Pharyngitis via a systematic "compound-target-pathway" analysis" is original, has not been published before and is not currently being considered for publication elsewhere.

We would like to draw the attention of the Editor to the following publications of one or more of us that refer to aspects of the manuscript presently being submitted. Where relevant copies of such publications are attached.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

CRediT authorship contribution statement

Qing Zhao: Writing – original draft, Investigation, Formal analysis, Conceptualization. **Xiaohuan Wang:** Validation, Data curation. **Yang Jin:** Formal analysis, Conceptualization. **Tiantai Wu:** Investigation, Formal analysis. **Ming Zhang:** Methodology. **Wanfen Shu:** Supervision. **Herong Li:** Visualization, Software. **Jian Wen:** Investigation. **Yan He:** Software, Resources. **Jiyu Chen:** Methodology, Conceptualization. **Shuaishuai Chen:** Investigation. **Ting Liu:** Software, Resources. **Jianfeng Zhang:** Supervision, Funding acquisition. **Wen Liu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper, except that Jianfeng Zhang is employed by Guizhou Valid Pharmaceutical Co., Ltd.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2025.156423](https://doi.org/10.1016/j.phymed.2025.156423).

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